Campaign Review & Approval Platform

ARCHITECTURAL SPECIFICATION & EXECUTIVE SUMMARY • DRAFT V1.0 • 01 NOVEMBER 2026

MAL4 Track: MarTech

The Problem

Marketing campaign proposals move across marketers, reviewers, and approvers before scheduling. When coordinated through disconnected tools or chats, teams lose state visibility, approve one version and schedule another, miss required revisions, or lack proof of who approved what. Concurrent edits and uncontrolled post-approval changes undermine brand compliance, budget authorization, and auditability.

Proposed Solution

A mobile-first, workflow-driven platform that manages campaign proposals from draft through review, revision, and approval, enforcing an automated scheduling gate. Every review decision is cryptographically bound to an immutable campaign version; only the approved version may be scheduled. Provides role-based actions, version history, and an append-only audit trail.

Core User Journey (Deterministic 6-Stage Gate)

1 DRAFT — CREATE
Marketer defines objective, audience, budget, schedule, and assets.

2 LOCK — SUBMIT
Validates required fields; locks immutable snapshot vN for review.

3 REVIEW — REVIEW
Reviewer inspects vN; approves, rejects, or requests changes.

4 LOOP — REVISE
Marketer updates draft based on feedback; resubmits as vN+1.

5 SIGNED — APPROVE
Explicit approval sign-off permanently bound to snapshot vN.

6 GATE — SCHEDULE
Gate verifies proposal matches approved version before release.

Architectural Approaches & Defensive Rationale (Chosen vs. Rejected Alternatives)

1. Explicit State Machine

Chosen: Deterministic finite state machine across 6 strict states.

Why Not BPMN Engines: Overkill for 6 states. Camunda/Temporal adds external failure points and runtime latency.

2. Immutable Snapshots

Chosen: Full payload snapshot locked at submission (`vN`).

Why Not In-Place Edits / Diffs: In-place mutation causes post-approval drift. Event diffs add projection latency on mobile.

3. Optimistic Concurrency

Chosen: Version-token / ETag verification (`version == current`).

Why Not Pessimistic Locks / 2PC: Network drops holding locks freeze mobile clients. 2PC creates distributed deadlocks.

4. Decoupled Async Outbox

Chosen: Atomic local state change + async retryable notifications.

Why Not Sync Inline Calls: Downstream push/email outages (FCM/SendGrid 500) would block legitimate approvals.

5. Server-Side RBAC & SoD

Chosen: Server-enforced role tokens with strict check (`creator != reviewer`).

Why Not Client Checks / Self-Review: Bypassed via cURL/API. Self-approval violates SOX 404 & FINRA 2210 compliance.

6. Append-Only Audit Log

Chosen: Chronological event ledger + structured JSON diagnostic logs.

Why Not In-Place Overwrites: Overwriting records destroys forensic audit proof needed for regulatory compliance.

IN SCOPE (SYSTEM CORE)
Campaign metadata, validation, submission, review gates, revision loop, immutable version snapshots, scheduling gate, RBAC, separation of duties, append-only audit trail.

OUT OF SCOPE (EXTERNAL ECOSYSTEM)
Creative authoring tools, audience segmentation engine, actual email/ad delivery execution, multi-touch attribution analytics, A/B testing, external CRM/DAM sync.

CORE PRINCIPLE: An approval belongs to one immutable campaign version; scheduling must verify that exact version. Architecture-led design for MAL4.